Detecting Dementia using Natural Language Processing

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***Abstract*—Dementia is a progressive neuro-degenerative disorder that affects memory, language, and cognitive skills. Early diagnosis is critical for effective intervention. This study investigates the use of Natural Language Processing (NLP) and machine learning models to detect dementia from speech transcripts. Multiple word representation techniques are explored, including TF-IDF, fastText embeddings, and BERT embeddings, along with four classifiers: Logistic Regression, SVM, Decision Tree, and Random Forest. Among the evaluated approaches, Decision Tree combined with fastText embeddings demonstrated superior performance on unseen test data, achieving a notably high F1-score. The findings highlight the potential of integrating deep semantic representations with classical machine learning techniques for effective dementia detection.**

***Index Terms*—Dementia Detection, Natural Language Processing (NLP), Machine Learning, BERT Embeddings, fastText, TF-IDF, ML Classifiers, Speech Transcript Analysis**

GitHub link to the project source code: https://github.com/nafis4139/Detecting-Dementia-using-Natural-Language-Processing-DAT550

I. Introduction

Dementia is a progressive neurological disorder characterized by a decline in cognitive functions, including memory, language, reasoning, and problem-solving abilities. It severely impacts a patient's ability to perform everyday activities independently and imposes significant emotional and financial burdens on families and healthcare systems. Early diagnosis of dementia is critical, as timely interventions can help manage symptoms, slow disease progression, and improve quality of life.

In recent years, Natural Language Processing (NLP) techniques have emerged as promising tools for analyzing linguistic patterns that may signal cognitive decline. Since language impairments often appear in the early stages of dementia, analyzing speech transcripts offers a non-invasive and cost-effective approach to early detection. Machine learning models, when combined with carefully extracted linguistic features, can differentiate between dementia patients and healthy individuals based on subtle language use patterns.

This project investigates the application of machine learning and NLP techniques to automatically detect dementia from speech transcripts. Several word representation methods, including TF-IDF, fastText embeddings, and BERT embeddings, are explored. Multiple classifiers such as Logistic Regression, Support Vector Machine (SVM), Decision Tree, and Random Forest are evaluated to determine the most effective approach. The study aims to highlight the potential of automated linguistic analysis in supporting early dementia diagnosis.

II. Literature Review

Detecting cognitive impairment through language analysis has attracted significant research interest in recent years. Traditional approaches often involved manual linguistic feature engineering, such as analyzing word frequency, syntactic complexity, or discourse coherence. With the rise of Natural Language Processing (NLP) and machine learning, more automated and scalable techniques have emerged.

Luz et al. (2020) introduced the ADReSS Challenge, a benchmark dataset and evaluation protocol for detecting Alzheimer's dementia from spontaneous speech, highlighting the potential of linguistic and acoustic feature extraction [1]. Fraser et al. (2016) demonstrated that NLP techniques could identify linguistic markers of Alzheimer's disease, using tasks such as picture description and narrative recall [2]. Their work showed that subtle language changes could be quantitatively measured long before traditional clinical symptoms appear.

More recently, Balagopalan et al. (2020) emphasized the application of deep learning models and transformer-based embeddings, such as BERT, for cognitive impairment detection [3]. Their study found that pre-trained language models, when fine-tuned on clinical data, could improve dementia classification performance compared to conventional machine learning approaches.

Despite these advances, challenges persist, including small dataset sizes, variability in spontaneous speech, and the need for interpretable models in clinical settings. This project builds upon these research efforts by systematically comparing TF-IDF, fastText, and BERT embeddings combined with multiple classifiers to detect dementia from speech transcripts.

III. Dataset Description

The dataset used in this project is the **Alzheimer's Dementia Recognition through Spontaneous Speech (ADReSS) Challenge 2020** subset from the DementiaBank corpus [4]. It consists of manually transcribed speech samples collected from both dementia patients and healthy controls during standardized tasks (e.g., picture description). The dataset was designed to be balanced and age-matched to minimize demographic bias in machine learning experiments.

Dataset Overview

- **Control_db:** 54 samples (Healthy individuals)
- **Dementia_db:** 54 samples (Dementia individuals)
- **Testing_db:** 48 samples (Hidden labels)

Each sample consists of metadata and a transcript of a picture description task. Each training file (`Control_db.csv` and `Dementia_db.csv`) contains 10 columns, while the testing file contains 9 columns (excluding the Diagnosis label):

- **Language**: Constant value "eng" for all samples.
- **Data**: Constant value "Pitt".
- **Participant**: Constant value "PAR".
- **Age**: Varies across participants (22 unique ages in Control, 21 unique ages in Dementia).
- **Gender**: Two categories: `male` and `female`.
- **Diagnosis**: In `Control_db`, always "Control"; in `Dementia_db`, four diagnosis types observed: "ProbableAD", "PossibleAD", "Memory", and "Vascular".
- **Category**: Binary class indicator (0 for Control, 1 for Dementia).
- **MMSE (Mini-Mental State Examination)**: Cognitive assessment scores (some missing values in the Control group).
- **Filename**: Unique identifiers for each sample.
- **Transcript**: The textual description produced by participants.

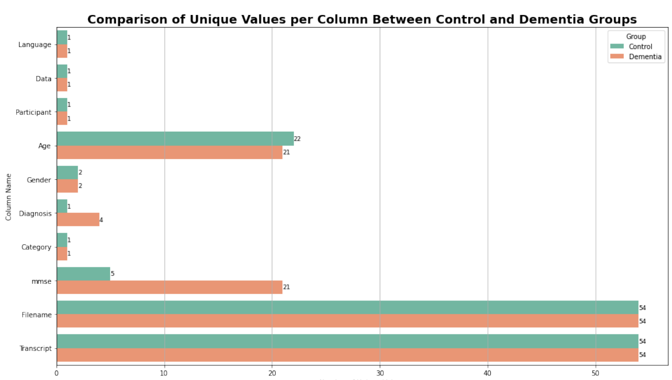

Fig. 1. Comparison of Unique Values per Column

In both Control and Dementia groups, the class distribution is balanced with 54 samples each (as shown in Figure 1). The test set includes 48 samples whose labels were used for final evaluation after model development. During preprocessing, missing MMSE scores were imputed based on the median MMSE score of the corresponding class group. Only selected columns (Age, Gender, MMSE, and Transcript) were utilized for feature extraction and modeling purposes, as the remaining columns were constant and provided no discriminative power.

IV. Data Visualization

To gain initial insights and better understand the distribution of both demographic and linguistic features, a set of exploratory data visualizations was performed. These visualizations help reveal important patterns and potential biases in the dataset.

A. MMSE Score Distribution

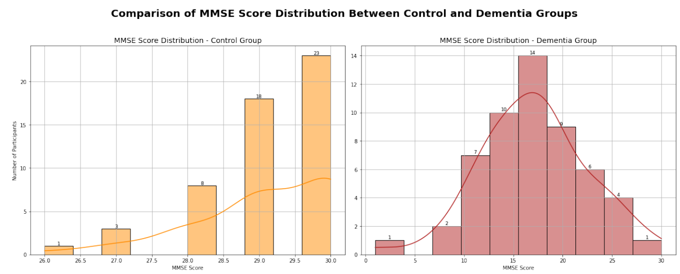

Fig. 2. MMSE score distributions: Control vs. Dementia

Figure 2 illustrates the MMSE score distributions for both Control and Dementia groups. The control group displays a tightly clustered distribution near the higher end of the MMSE scale (typically scores $\geq$27), indicating generally preserved cognitive function. In contrast, the dementia group exhibits a broader and more left-skewed distribution, with many individuals scoring below 25, which reflects varying levels of cognitive impairment.

This distribution confirms that MMSE is a strong clinical indicator of cognitive status and supports its inclusion as a critical numerical feature in the machine learning models.

B. Transcript Length Distribution

Figure 3 compares the distribution of transcript lengths between dementia and control participants. The histogram shows that dementia patients tend to produce shorter transcripts, with a clear right-skewed distribution concentrated in the 45–90 word range. This pattern likely reflects reduced linguistic fluency and cognitive retrieval issues.

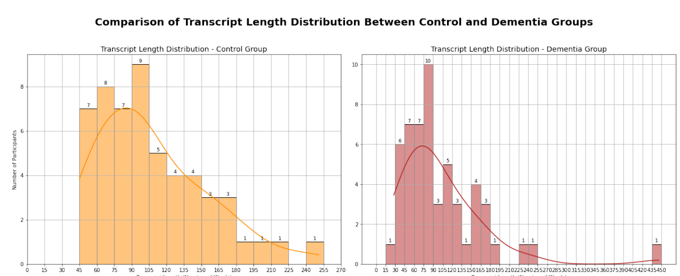

Fig. 3. Transcript length distribution: Control vs. Dementia

In contrast, the control group demonstrates a more symmetric and wider distribution, with several participants generating transcripts longer than 150 words. This suggests more elaborate and fluent speech typical of healthy individuals. These patterns confirm that transcript length is a meaningful feature for distinguishing between the two groups. The distribution range is also summarized below:

- **Short Responses (15–75 words)**: Dementia participants dominate, especially in the 15 to 60 word ranges.
- **Mid-range (75–135 words)**: Both groups are represented, though control cases are slightly more concentrated.
- **Long Responses (>150 words)**: More frequent in the control group, indicating higher expressive capacity.

C. Age & Gender Distribution

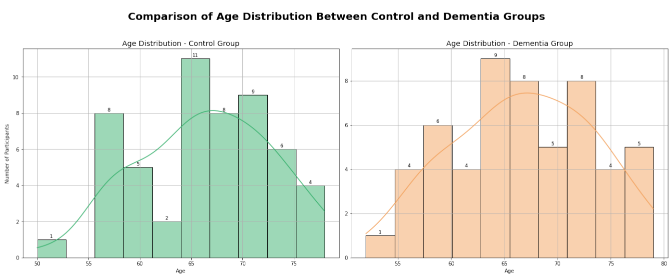

Fig. 4. Age distribution: Control vs. Dementia

The age distribution plot in figure 4 reveals that both the Control and Dementia groups span a similar age range. However, the Control group appears to cluster more tightly around a slightly higher mean, whereas the Dementia group exhibits a broader, more dispersed distribution. This suggests that age, while relevant, may not be a strong standalone predictor for dementia classification. Gender distribution (figure 5) is balanced across classes, reducing the risk of model bias and supporting a fair evaluation framework.

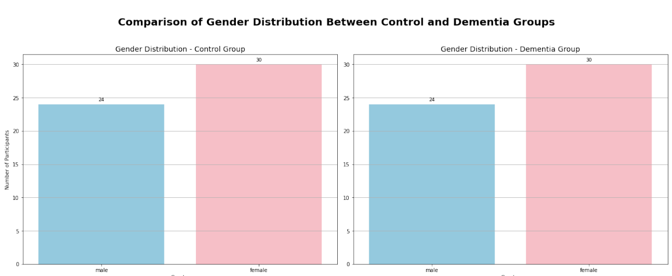

Fig. 5. Gender distribution: Control vs. Dementia

V. Data Preprocessing

Before training the machine learning models, several preprocessing steps were performed to clean and structure the dataset effectively for feature extraction and classification tasks. The major preprocessing steps included:

- **Transcript Cleaning:** Non-verbal annotations such as noises (`[noise]`, `[//]`), tags (`<tag>`), and fillers (e.g., `&uh`, `&um`) were removed using regular expressions. Punctuation marks were stripped, and text was converted to lowercase to standardize the corpus.
- **Tokenization and Lemmatization:** Each cleaned transcript was tokenized into individual words. Words were then lemmatized to their base forms using the WordNet lemmatizer, helping to reduce inflectional variations (e.g., "running" $\rightarrow$ "run").
- **Optional Stopword Removal:** Stopwords (common words such as "the," "and," "is") were optionally removed to focus on meaningful content words during initial experiments. However, for deep learning models such as BERT, stopwords were retained to preserve context.
- **Handling Missing MMSE Scores:** In case of any missing values in the MMSE field, a function were imputed using the median MMSE score within each diagnosis group (dementia or control) to fill in the missing values.
- **Encoding Gender Information:** Gender, originally a categorical variable (male or female), was numerically encoded (0 for male, 1 for female).
- **Additional Feature Engineering:** Additional numerical features such as the total transcript length (number of words) were computed for each sample to capture speech verbosity, which can be an important marker of cognitive impairment.

Through this preprocessing pipeline, both linguistic and demographic features were prepared systematically, enabling the application of different feature extraction methods and classification models.

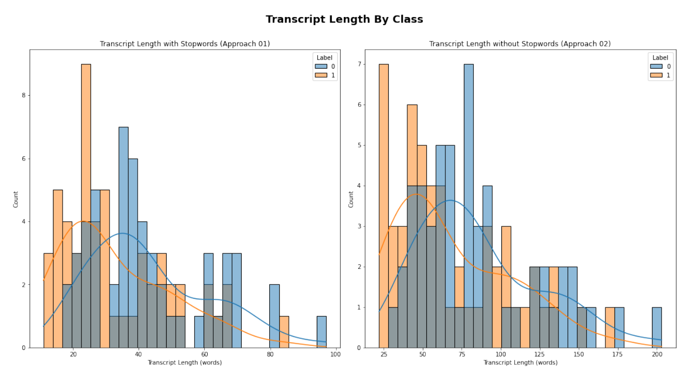

Fig. 6. Comparison of Transcript Length Distributions between Two Preprocessing Approaches

The comparison in figure 6 shows that removing stopwords shortens transcripts and highlights the linguistic gap between dementia and control groups—dementia transcripts are noticeably shorter. While this benefits traditional models by emphasizing differences, retaining stopwords is better for deep learning models like BERT, which rely on full context. Since this project uses BERT alongside fastText and TF-IDF, stopwords were retained to preserve contextual meaning.

VI. Feature Extraction

To enable machine learning models to effectively process textual data, three different word representation techniques were explored in this project: **TF-IDF**, **fastText embeddings**, and **BERT embeddings**. These methods capture different levels of semantic and syntactic information from the transcripts.

A. TF-IDF (Term Frequency-Inverse Document Frequency)

TF-IDF is a classical statistical measure that evaluates the importance of a word within a document relative to a collection of documents [5]. In this project, TF-IDF was used to convert cleaned speech transcripts into numerical vectors:

- A `TfidfVectorizer` with a maximum of 1000 features was applied to the preprocessed text.
- This generated a sparse matrix representing the importance of terms across the dataset.
- Numeric features such as MMSE score, transcript length, gender, and age were standardized using `StandardScaler` and concatenated with the TF-IDF vectors.
- The final combined feature set had 685 dimensions (681 from TF-IDF and 4 from numeric features).
- The dataset was split into 80% training (86 samples) and 20% validation (22 samples) for model evaluation.

This approach balances linguistic and demographic indicators, providing a meaningful feature representation for training machine learning classifiers.

B. fastText Embeddings

fastText is a word embedding model developed by Facebook AI Research that maps words to high-dimensional continuous vectors using subword information [6]. In this project, fastText was used to generate fixed-length vector representations for each transcript:

- Each word in the transcript was mapped to its corresponding 300-dimensional fastText embedding.
- For each transcript, word embeddings were averaged to form a single fixed-length vector representation.
- These averaged vectors were concatenated with the structured numeric features.
- To reduce memory usage, a trimmed version of the fastText embedding set was created containing only the words present in the training corpus.

The resulting feature set comprised 304 dimensions: 300 from fastText and 4 from the structured numeric features. The dataset was then split into training (86 samples) and validation (22 samples) using an 80-20 stratified split. This combination of semantic and numeric data provided a compact yet informative representation suitable for traditional classifiers.

C. BERT (Bidirectional Encoder Representations from Transformers) Embeddings

BERT is a transformer-based deep learning model that captures the contextual meaning of language by considering both the left and right context of words [7]. In this project, BERT was leveraged to encode the semantic content of each transcript as follows:

- The pre-trained `bert-base-uncased` model from Hugging Face Transformers was used.
- For each transcript, the 768-dimensional [CLS] token embedding was extracted, representing the entire utterance's contextual representation.
- These embeddings were combined with structured numeric features: MMSE score, transcript length, gender, and age.

The resulting feature matrix had 772 columns—768 from BERT and 4 from the numeric data. This enriched representation was then split into training (86 samples) and validation (22 samples) sets using a stratified 80-20 split. The use of BERT embeddings provided a powerful, modern representation that complements traditional approaches and is especially suited for downstream classification tasks involving complex language patterns.

VII. Model Training

To evaluate the effectiveness of different text representations in detecting dementia, four classical machine learning classifiers were trained:

- Logistic Regression
- Support Vector Machine (SVM)
- Decision Tree
- Random Forest

Each model was trained independently on three distinct feature sets:

- **TF-IDF + Numeric:** Bag-of-words representation combined with MMSE, age, gender, and transcript length.
- **fastText + Numeric:** Average word embeddings combined with numeric features.
- **BERT + Numeric:** [CLS] token embedding representing full transcript semantics, concatenated with numeric features.

An 80-20 train-validation split was applied to ensure robust evaluation across all combinations. For each classifier-feature pair, key evaluation metrics were computed: Accuracy, Precision, Recall, F1-score and ROC AUC. All models were implemented using `scikit-learn`. Evaluation results were stored for comparative analysis and used to generate visualizations. To better understand model behavior across representations, ROC curves and confusion matrices were plotted:

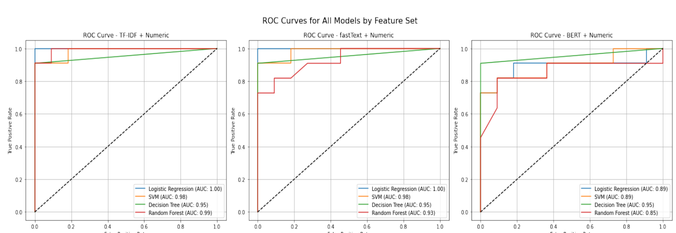

Fig. 7. ROC Curves for All Classifiers (Train Set)

- **ROC Curves:** Plotted for all models across each feature type (figure 7). These curves illustrate the trade-off between sensitivity and specificity.
- **Confusion Matrices:** Heatmaps (figure 8) displayed correct and incorrect classifications. These plots highlighted areas where models frequently confused dementia and control samples, offering insight into classifier reliability.

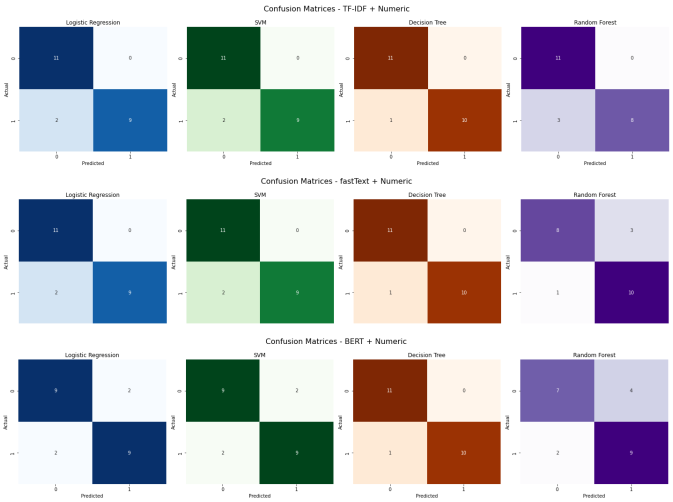

Fig. 8. Confusion Matrices for All Classifiers (Train Set)

This systematic evaluation across different combinations of embeddings and classifiers enabled the selection of the best-performing pipeline for test set validation.

VIII. Results and Evaluation

This section presents the performance of the trained models using different feature representations: TF-IDF, fastText, and BERT embeddings. Evaluation was conducted on both validation and unseen test sets using standard metrics: Accuracy, Precision, Recall, F1 Score, and ROC AUC.

A. Training and Validation Performance

Table I summarizes the validation results sorted by F1 Score. Decision Tree classifiers using TF-IDF, fastText, and BERT all achieved the highest F1 score of 0.95. Logistic Regression and SVM performed comparably well with TF-IDF and fastText, suggesting that traditional text representations still hold strong predictive power. BERT-based models performed reasonably but showed slightly lower F1 scores during training, potentially due to smaller dataset size.

Model	Accuracy	Precision	Recall	F1	ROC AUC
Decision Tree (TF-IDF)	0.95	1.00	0.91	0.95	0.95
Decision Tree (fastText)	0.95	1.00	0.91	0.95	0.95
Decision Tree (BERT)	0.95	1.00	0.91	0.95	0.95
Logistic Regression (TF-IDF)	0.91	1.00	0.82	0.90	1.00
SVM (TF-IDF)	0.91	1.00	0.82	0.90	0.98
Logistic Regression (fastText)	0.91	1.00	0.82	0.90	1.00
SVM (fastText)	0.91	1.00	0.82	0.90	0.98
Random Forest (TF-IDF)	0.86	1.00	0.73	0.84	0.99
Random Forest (fastText)	0.82	0.77	0.91	0.83	0.93
Logistic Regression (BERT)	0.82	0.82	0.82	0.82	0.89
SVM (BERT)	0.82	0.82	0.82	0.82	0.89
Random Forest (BERT)	0.73	0.69	0.82	0.75	0.85

TABLE I
Validation Performance of All Models (Sorted by F1 Score)

B. Test Set Evaluation

Table II presents the evaluation results on the held-out test set. The highest F1 score of 0.91 was jointly achieved by Decision Tree with fastText features and Logistic Regression with BERT embeddings. This demonstrates that both semantic embeddings (BERT, fastText) and classical interpretable models (Decision Tree, Logistic Regression) can be effectively leveraged for detecting dementia.

Model	Accuracy	Precision	Recall	F1 Score	ROC AUC
Decision Tree (fastText)	0.92	0.95	0.88	0.91	0.92
Logistic Regression (BERT)	0.92	0.95	0.88	0.91	0.94
Decision Tree (BERT)	0.90	0.91	0.88	0.89	0.90
Decision Tree (TF-IDF)	0.90	0.95	0.83	0.89	0.90
Logistic Regression (TF-IDF)	0.90	1.00	0.79	0.88	0.98
Logistic Regression (fastText)	0.90	1.00	0.79	0.88	0.98
Random Forest (TF-IDF)	0.85	0.90	0.79	0.84	0.93
SVM (fastText)	0.85	1.00	0.71	0.83	0.94
Random Forest (fastText)	0.83	0.86	0.79	0.83	0.95
Random Forest (BERT)	0.83	0.90	0.75	0.82	0.94
SVM (TF-IDF)	0.83	0.94	0.71	0.81	0.96
SVM (BERT)	0.83	0.94	0.71	0.81	0.94

TABLE II
Test Set Performance of All Models (Sorted by F1 Score)

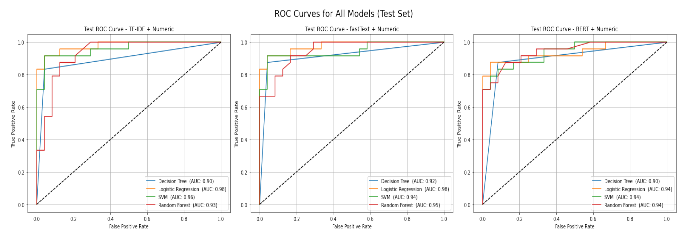

Fig. 9. ROC Curves for All Classifiers (Test Set)

a) ROC Curves: Figure 9 presents the Receiver Operating Characteristic (ROC) curves for all classifiers on the test set. A well-performing model will have a curve that bows toward the top-left corner of the plot, indicating a high true positive rate and a low false positive rate. The Logistic Regression model with BERT embeddings achieved the highest Area Under the Curve (AUC = 0.94), followed closely by models based on fastText and TF-IDF features. This suggests that semantic embeddings, particularly when combined with linear classifiers, are highly effective at distinguishing between dementia and control cases.

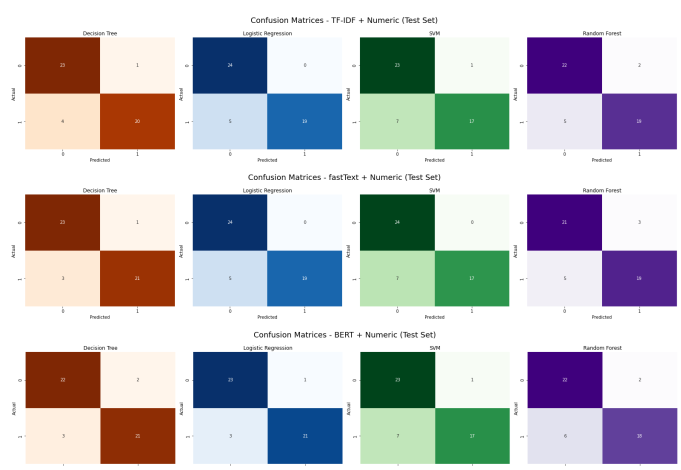

Fig. 10. Confusion Matrices for All Classifiers (Test Set)

b) Confusion Matrices: Figure 10 shows the confusion matrices for each classifier evaluated on the test set. Most models demonstrate a high number of correctly classified instances in both dementia and control categories. Decision Tree and Logistic Regression models, especially when using fastText or BERT features, exhibit strong diagonal dominance—indicating high accuracy and balanced performance across classes. False positives and false negatives remain minimal, supporting the reliability of the models for practical use.

C. Insights

- Decision Trees performed consistently well across all feature sets, particularly in fastText and BERT configurations.
- Logistic Regression with BERT generalizes better on test data despite slightly lower training performance.
- TF-IDF features paired with classical models (LR, SVM) still offer competitive results.
- ROC curves and confusion matrices illustrate strong true positive rates and minimal false classifications.

IX. Limitations and Future Work

Despite the encouraging results obtained in this study, several limitations must be acknowledged, and they point to valuable directions for future research:

- **Limited Dataset Size:** The ADReSS dataset is relatively small, with only 108 labeled training samples. This constrains the complexity of models that can be effectively trained without overfitting. Future studies could benefit from larger, more diverse datasets covering different stages and types of dementia.
- **Text-Only Analysis:** This project focuses exclusively on transcript data and does not incorporate acoustic or paralinguistic features such as pitch, pauses, or speech rate. These audio characteristics are known to correlate with cognitive decline and could enhance classification accuracy when used alongside text features.
- **No Longitudinal Tracking:** The dataset includes only single-session interviews per participant. Longitudinal speech data collected over time would allow for more accurate monitoring of cognitive decline and could support early detection.
- **Feature Engineering Scope:** Although several textual and numeric features were considered, additional linguistic cues such as part-of-speech ratios, syntactic complexity, and lexical richness were not explored in depth. Future work could include a broader set of linguistic indicators.
- **Pre-trained Embedding Limitations:** While fastText and BERT provide strong baselines, they are trained on general English corpora and not specifically adapted to clinical or elderly speech. Domain-adapted embeddings or fine-tuning on similar medical text could improve their effectiveness.
- **Model Interpretability:** Deep learning models like BERT offer powerful representations but can be difficult to interpret. For practical deployment in clinical settings, methods that offer transparency and explainability (e.g., SHAP or LIME) should be explored.

Addressing these limitations can help advance the field of speech-based dementia detection and lead to more robust, generalizable, and clinically applicable tools.

X. Conclusion

This project explored the application of Natural Language Processing (NLP) and machine learning techniques to the early detection of dementia from speech transcripts. Three types of text feature representations—TF-IDF, fastText, and BERT—were investigated in combination with four machine learning classifiers: Logistic Regression, SVM, Decision Tree, and Random Forest. These models were evaluated using a range of performance metrics including F1 score, ROC AUC, and confusion matrices.

The experimental results demonstrated that both classical and embedding-based features can effectively distinguish between dementia and control cases. Decision Trees consistently performed well across all feature types, while Logistic Regression paired with BERT embeddings offered the best generalization on unseen test data. TF-IDF features, although simpler, also provided strong performance with low computational cost.

Overall, the findings highlight that carefully engineered linguistic and numerical features, when combined with robust classifiers, can contribute meaningfully to automated dementia detection. The study reinforces the potential of NLP-based screening tools in clinical and assistive healthcare applications.

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